



उत्पाद मैनुअल

आई एस 3854 : 2023 के अनुसार

घरेलु और समान कार्यों के लिए स्विच - विशिष्टि (तीसरा पुनरीक्षण)के लिए

दस्तावेज संख्या - पीएम/आईएस 3854/3/ जनवरी 2024

विभिन्न उत्पादों के लिए भारतीय मानक ब्यूरो (अनुरूपता मूल्यांकन) विनियम, 2018 की योजना- I के तहत प्रमाणन के संचालन में एकरूपता और पारदर्शिता के लिए इस उत्पाद मैनुअल का उपयोग सभी क्षेत्रीय / शाखा कार्यालयों और लाइसेंसधारियों द्वारा संदर्भ सामग्री के रूप में किया जाएगा। दस्तावेज का उपयोग बीआईएस प्रमाणन प्राप्त करने के इच्छुक संभावित आवेदकों द्वारा भी किया जा सकता है।

PRODUCT MANUAL FOR SWITCHES FOR DOMESTIC AND SIMILAR PURPOSES — SPECIFICATION (THIRD REVISION) ACCORDING TO IS 3854 : 2023

Document No - PM/IS 3854/3/January 2024

This Product Manual shall be used as reference material by all Regional/Branch Offices & licensees to ensure uniformity of practice and transparency in operation of certification under Scheme-I of Bureau of Indian Standards (Conformity Assessment) Regulations, 2018 for various products. The document may also be used by prospective applicants desirous of obtaining BIS certification.

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PRODUCT MANUAL
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1.	मानक संख्या IS No.	:	IS 3854 : 2023
	शीर्षक Title	:	Switches for Domestic and Similar Purposes — Specification (Third Revision)
	संशोधनों की संख्या No. of amendments	:	0
2.	नमूना दिशानिर्देश Sampling Guidelines:		
a)	कच्चा माल Raw material	:	NA
b)	समूहीकरण दिशानिर्देश Grouping Guidelines	:	Please refer ANNEX – A

c)	नमूने का परिमाण Sample Quantity	:	25 Switches
3.	परीक्षण उपकरणों की सूची List of Test Equipment	:	Please refer ANNEX – B
4.	निरीक्षण और परीक्षण की स्कीम Scheme of Inspection and Testing	:	Please refer ANNEX – C
5.	एक दिन में संभावित परीक्षण Possible tests in a day:	:	Please refer ANNEX-D
6.	लाइसेंस का दायरा /Scope of the Licence: Please refer ANNEX-E		
7.	अन्य दिशानिर्देश/ Any other guidelines: Please refer ANNEX-F for guidelines for outsourcing of moulded bodies		

ANNEX – A

समूहीकरण दिशानिर्देश Grouping Guidelines

1. The following aspects of a switch shall be taken into consideration for grouping of different varieties of switches for the purpose of GoL/ CSoL:
 - a. Current rating
 - b. Voltage rating
 - c. Possible connections or pattern
 - d. Amperes rating (rated current for externally ballasted loads)
 - e. Contact opening
 - f. Degree of protection against harmful ingress of solid foreign objects
 - g. Degree of protection against harmful ingress of water
 - h. Method of activating the switch
 - i. Method of mounting the switch
 - j. Type of terminals
 - k. Degree of protection against electric shock
2. To cover a range of current ratings, switches from the minimum and maximum rating of that range shall be tested.
3. The following relaxation may be given when a variety is tested for all the requirements:
 - a) If a particular voltage rating is tested, lower ratings may also be covered.
 - b) If a pattern is tested, additions pattern may be covered as given below:

Pattern tested	Additional Pattern
6	1
4	6,1
6/2	6,2
7	6/2, 2
03	3

- c) If switch with Amperes rating (AX) (rated current for externally ballasted loads) is tested, then separate testing may not be carried out for normal current rating (A).
 - d) If switch with a particular degree of protection is tested, switches with lower degree of protection may be covered.
4. For all other aspects mentioned at 1.1(h) to 1.1(j) above, each aspect shall be tested to cover the entire range.

5. The Firm shall declare the varieties of Switches intended to be covered in the Licence.
6. Scope of licence shall be restricted based on the manufacturing and testing facilities available.
7. During operation of licence, samples of each variety covered in the scope of licence, shall be tested in rotation, to the extent possible.

ANNEX – B

परीक्षण उपकरणों की सूची

List Of Test Equipment***(INDICATIVE LIST, FOR GUIDANCE ONLY)******Major test equipment required to test as per the Indian Standard***

Sl. No.	Tests Used In With Clause Reference	Test Equipment/Chemical
1.	Marking - 8.9	Petroleum Spirit
2.	Dimensions - 9	Vernier Calliper
3.	Protection against electric shock - 10	Un-jointed test finger, Standard test finger, Voltmeter , Voltage Source with indicator
4.	Earthing Resistance - 11.4	Low voltage source with Voltmeter and ammeter
5.	Resistance to Humidity - 15.3	Humidity Controlled Cabinet, Thermometer Hygrometer
6.	Insulation resistance - 16.1	Megger (500 V dc)
7.	Electric strength (flash test) - 16.2, 16.3	High voltage Tester (ac), Voltmeter, Ammeter
8.	Resistance to Ageing, Heat, Rusting - 15, 21, 25	Electrical Oven
9.	Temperature Rise Test, Making-Breaking and Normal Operation Test - 17, 18, 19	Temperature Rise Test Equipment, Thermometer(Digital), Ammeter, Pine wood block
10.	Terminals - 12, 13, 22	Torque Screw Driver
11.	Making-Breaking Capacity and Normal operation test – 18, 19	Test Apparatus, cycle counter, ON Timer OFF Timer, Inductive Load of required power factor, Voltmeter, Auto Transformer, PF Meter, Tungsten Filament Lamp load, Fluorescent Lamp Load
12.	Mechanical Strength – 20	Mechanical Strength (Impact test apparatus) with plywood base for mounting (Pendulum Type)
13.	Pendulum Hammer Test - 20.2	Cylinder of steel sheet and flat steel sheet (for surface type switches)

14.	Test on the Main Parts of Surface-type Switches - 20.3	Cylindrical metal rod
15.	Screwed Glands - 20.4	Setup for verification of removal and non-removal of covers
16.	Covers, Cover plates - 20.7, 20.8	Gauges as per Fig 18 to 21
17.	Grooves, Holes and Reverse Tapers - 20.9	Test setup on cord operated switch
18.	Resistance to heat – 21	Ball Pressure Apparatus (Fig 23), Magnifying glass
19.	Resistance to Rusting – 25	CCL ₄ , NH ₄ Cl, Distilled Water & Beaker
20.	Resistance to Ageing - 15.1	Weighing Balance, Weights
21.	Protection provided by enclosures of switches -15.2	Splash proof / Jet proof test setup (for IPX4/IPX5)
22.	Resistance to Abnormal Heat, Fire and Tracking - 24	Glow Wire Test Apparatus with Temperature Indicator & Controller, Timer pre-set Time Interval Indicator, Tracking test apparatus (for switches other than ordinary)
23.	Terminals & Screws - 12.2.5	Test setup for checking conductor damage with rotational speed controller and weights
24.	Screw less Terminal testing - 12.3.10, 12.3.11, 12.3.12	Test setup for mechanical stress, electrical and thermal stress, and Deflection test on Screw less terminals
25.	Attachment of knobs - 13.5	Testing arrangement for knob of rotary switches
26.	Requirements for Membranes in Inlet Openings - 13.15	Fixture for membrane reliability test
27.	Pull force test on cord operate switch - 14.6	Pull force test on cord operate switch setup
28.	Test for Switches Intended for Self-Ballasted Lamp Loads – 19.3	Load bank

Note: The above list is indicative only and may not be treated as exhaustive.

ANNEX – C

निरीक्षण और परीक्षण की स्कीम

Scheme Of Inspection And Testing

1. QUALITY ASSURANCE PLAN

1.1 It is expected that manufacturers (licensees/applicants) will implement a Quality Assurance Plan i.e. a plan of regular testing and in-process controls, designed to ensure that the product bearing the Standard Mark conforms to all requirements of the Indian Standard.

1.2 The manufacturers shall define a Quality Assurance Plan defining the control unit (i.e. lot/batch etc.) and the levels of control (i.e. the frequency and number of samples for conducting the different tests as per the Indian Standard) and submit the same to BIS Branch Office for information. The manufacturer shall comply with the same and maintain test records in accordance with para 2.4.

1.3 RECOMMENDED LEVELS OF CONTROL/CONTROL UNIT:

1.3.1 For the guidance of manufacturers, the recommended definition of control unit is : Switches of same type and rating manufactured in a day shall constitute a control unit. The manufacturer has the choice to define their own control unit/batch/lot and submit the same to the BIS

1.3.2 For the guidance of manufacturers in preparing the Quality Assurance Plan, recommended levels of control are given in **Table 1**.

1.3.3 The manufacturer shall ensure inspection and testing as per the Quality Assurance Plan submitted by them on the whole production of the factory which is covered by this plan. Alternatively, the manufacturer has the option of adherence to the quality plan as per levels of control recommended in column 3 of Table 1.

1.4 However, all manufacturers shall ensure compliance of their products to all the requirements of the Indian Standard.

2. ENSURING COMPLIANCE THROUGH TESTING- It is expected that manufacturers (licensees/applicants) will establish a suitably equipped and staffed in house laboratory (In house testing facility) for testing at least those parameters of the Indian Standard which require routine testing for ensuring quality of the product. This includes in-process controls as may be defined and put in place by the manufacturer and testing parameters/requirements which can only be performed in the factory.

2.1 For the guidance of manufacturers, Table 1 giving the recommended levels of control is given below. Column 2 of Table 1 indicates routine tests where test equipment is required in house as “R” or other tests which can be subcontracted as “S”. Subcontracting is permitted to BIS recognized/empanelled laboratory or any other laboratory having valid NABL accreditation as per IS/ISO/IEC 17025.

2.2 For MSME manufacturers, the requirement of maintaining a laboratory/in-house testing facility for routine tests (indicated as “R” in Column 2 of Table 1) is also optional.

2.2.1 MSME manufacturers may utilize common cluster based facilities as per guidelines for the utilization of cluster based test facilities by MSMEs or the provisions of Sharing of testing facilities or get testing done from

BIS recognized/empaneled laboratory or any other laboratory having valid NABL accreditation as per IS/ISO/IEC 17025.

2.3 Large Scale manufacturers shall maintain an in-house laboratory equipped at least with test facilities for routine tests (indicated as “R” in Column 2 of Table 1), where different tests given in the specification shall be carried out in accordance with the method given in the specification. They shall also implement a calibration plan for the in-house test equipment.

2.3.1 Alternatively, in lieu of an in-house laboratory, large scale manufacturers can also utilize the provisions of Sharing of testing facilities as per the Guidelines for Grant of Licence available on BIS website www.bis.gov.in. (Under Conformity Assessment>Product Certification Process). Even for subcontracted tests, provisions for sharing of testing facilities can be utilized.

2.4 TEST RECORDS- The manufacturers maintaining an in-house laboratory or utilizing common cluster based facilities or shared test facilities shall maintain test records for the tests carried out to establish conformity. For the tests being subcontracted to BIS recognized/empanelled laboratory or any other laboratory having valid NABL accreditation as per IS/ISO/IEC 17025, test reports issued by the laboratories shall be available for inspection by BIS.

3. PACKING AND MARKING - The Standard Mark as given in the Schedule of the licence shall be incorporated legibly and indelibly on **each switches and also on its packaging** provided always that the material so marked conforms to each requirement of the specification.

3.1 Packing and Marking shall be done as per the **IS 3854 : 2023**.

3.2 Additional Marking requirements: The material shall also be marked with the following additional requirement on packaging:

a) “For BIS certification details please visit www.bis.gov.in”

4. REJECTION - All the production which conforms to the Indian Standard and covered under the scope of this licence shall be marked with the Standard Mark. Disposal of non-conforming product shall be done in such a way so as to ensure that there is no violation of provisions of BIS Act, 2016.

TABLE 1
(ONLY FOR GUIDANCE PURPOSE)

(1)				(2)	(3)		
Test Details				Test equipment requirement R: required (or) S: Sub-contracting permitted	Levels of Control		
Cl.	Requireme nt	Test Methods			No. of Sample	Frequency	Remarks
		Claus e	Referenc e				
8	Marking (except 8.9)	8	IS 3854	R		Each Switch	
16.3	Electric Strength (Flash test)	16.3		R	3	Every week for each type and rating	
8.9	Marking – Durability	8.9		R	3	Each control unit	
9	Checking of Dimensions	9		R	3	Each control unit	Refer Note 1
10	Protection against electric shock	10		R	3	Every week for each type and rating	
11	Provision for earthing	11		S		Once in three months for each type and rating	Refer Note 2
12	Terminals	12		S			
13	Constructional requirements	13		S			
14	Mechanism	14		S			
15	Resistance to ageing, harmful ingress of water and to humidity	15		S			
16	Insulation resistance and electric strength (except 16.3)	16		S			
17	Temperature rise	17		S			
19	Normal operation	19		S			
20	Mechanical Strength	20		S			

21	Resistance to heat	21		S			
22	Screws, current-carrying parts and connections	22		S			
23	Creepage distances, clearance and distances through sealing compound	23		S			
25	Resistance to rusting	25		S			
24	Resistance to insulating material to abnormal heat, to fire and to tracking	24		S	3	Once in a year or whenever there is change in raw material for each type and rating	Refer Note 2
18	Making and breaking capacity	18		S	3	Once in three years for each type and rating for each type and rating	Refer Note 2

Note-1: Firm should exercise adequate control so that whole production should meet the requirements. Records shall be maintained.

Note-2: The number of specimens for testing shall be as per selected as per clause 5.3 and Table 1 of IS 3854. In case of any failure, the provisions given in clause 5.3 of IS 3854 shall be followed.

Note-3: Sub-contracting is permitted to a laboratory recognized by the Bureau or Government laboratories empaneled by the Bureau

Note-4: Levels of control given in table above are only recommendatory in nature. The manufacturer may define the control unit/batch/lot and submit his own levels of control with proper justification for approval by BO Head.

ANNEX – D

एक दिन में संभावित परीक्षण Possible Tests in a day

1. Marking (Cl. 8)
2. Protection against electric shock (Cl. 10)
3. Electric Strength (Flash test) (Cl. 16.3)
4. Temperature rise (Cl. 17)
5. Making and breaking capacity – General (Cl. 18.1)
6. Mechanical Strength (Cl. 20)
7. Resistance to heat (Cl. 21)
8. Screws, current-carrying parts and connections (Cl. 22)
9. Creepage distances, clearance and distances through sealing compound (Cl. 23)
10. Resistance to rusting (Cl. 25)

ANNEX – E

लाइसेंस का दायरा
Scope of Licence

Licence is granted to use Standard Mark as per IS 3854:2023 with the following scope:

Name of the Product	Switches for domestic and similar purposes
Rated Current	
Rated Voltage	
Nature of supply	
Possible Connections (PatternNumber)	
Amperes rating (rated current for externally ballasted loads)	AX/A
Contact opening	Normal gap construction / mini-gap Construction/ micro-gap construction/ switches without contact gap (semiconductor switching devices)
Degree of protection against Electric Shock	Unenclosed / Enclosed
Degree of protection against harmful ingress of Water and against harmful ingress of solid foreign objects	IP....
Method of activating the switch	Rotary/ Tumbler/ Rocker/ Push-Button/ Cord-operated/ Dolly type
Method of Mounting	Surface-type/ Flush-type/ Semi flush type/ Panel-type/ Architrave-type
Type of Terminals	Screw-type terminals/ Screwless terminals for rigid conductors only/ Screwless terminals for rigid and flexible conductors only

ANNEX – F

Guidelines for Outsourcing of Moulded Bodies

a) Outsourcing of moulding facilities may be permitted with the following organizations (Moulder):

For Domestic Manufacturers

- Any dedicated moulding unit in the conforming/ authorized industrial area and located under the jurisdiction of the same Branch Office.
- Any other unit of the Licensee/ Applicant or their sister concern, which is having a valid BIS licence for the same product under the jurisdiction of the same Branch Office.

For Foreign Manufacturers

- Any dedicated moulding unit in the conforming/ authorized industrial area and located in the same city where the applicant/ licensee is located.
- b) The dedicated moulder shall not carry out moulding for units other than the particular Applicant/ Licensee.
- c) The Manufacturer shall take the following actions:
- Enter into an agreement with the moulder which shall comprise, inter alia, the terms and conditions for moulding including the condition of dedicated moulding, period of agreement etc.
 - Declare the Items and Brands for which he intends to outsource moulding.
 - Declare the name, ownership details, address, details of the responsible person, phone number, etc. of the moulder.
- d) It shall be the responsibility of the licensee (the firm which is outsourcing the moulding activity) to ensure that no finished material is produced or assembled at any place other than the licensed premises and he shall be responsible for any misuse of ISI mark at the moulder's premises. An Undertaking to this effect shall be furnished to BIS.
- e) Records of raw material issued date-wise, moulded material received back date-wise and item-wise shall be maintained by the licensee. Similar records shall also be maintained by the moulder.
- f) The moulder shall ensure that there is a separate dedicated space for keeping moulded components as well as the raw materials received from the licensee. Colour coding of bins/ separate storage location etc. may be used to ensure easy and quick traceability. Appropriate records are to be maintained at both the units.
- g) The licensee shall ensure that there are adequate means of ensuring the quality of incoming outsourced moulded bodies.

- h) For applications received with request for outsourcing of moulding facilities, a special inspection of the moulder's premises shall be carried out to adjudge the capability of the moulder in addition to the preliminary inspection at the applicant's premises. Special Inspection Charges and any other charges, as applicable, for this visit shall be collected from the applicant in advance. The process flow chart submitted by the applicant shall clearly indicate the outsourcing of moulded bodies. The recommendations for GoL shall include details on outsourcing of moulding operation by the firm. The details of outsourcing may also be included in the GoL intimation letter.
- i) For existing licensees who intend to outsource the moulding facility, the licensee shall submit a fresh process flow chart indicating the outsourcing of moulded bodies. The permission for outsourcing may be given to the licensee after complying with all the procedures including verification visit to the moulder's premises.
- j) Whenever there is a change in the moulder, the licensee has to enter into a fresh agreement with the new moulding unit. The permission for outsourcing may be given to the licensee after complying with all the procedures including verification visit to the moulder's premises.
- k) During the operation of the licence, records required to be maintained at the licensee end with respect to outsourced moulding shall be verified during surveillance visit at the licensee's premises by BIS Certification Officer.
- l) BIS Officer(s) shall have access to the moulder's premises and if at any time the same is denied, permission for outsourcing shall be withdrawn. An undertaking in this respect shall be obtained.
- m) For Domestic manufacturers, the visit to moulder's premises may be taken up as a special visit as and when needed. Special Inspection Charges for this visit shall be collected from the licensee after the visit. An undertaking from the licensee regarding payment of such charges shall be obtained.
- n) For Foreign manufacturers, BIS shall maintain surveillance over the moulder's premises in addition to the manufacturer's premises. Special inspection charges and all other applicable charges for this visit shall be payable by the licensee.

Note:

- The Indian Standard does not impose any restriction on the manner/ method of Marking so long as the Durability clause is complied with. As such, the product may be marked with the Standard Mark, CM/L Number, Manufacturer's Brand and other markings during the process of moulding itself, in which case all the conditions stated above shall apply.
- However, if the Marking on the moulded bodies is done at the Licensee's own premises by other means such as pad printing/ laser marking etc, the moulded component may be treated as any other bought out component/ raw material as these are not marked with the ISI Mark or CM/L Number during the outsourced moulding process. Notwithstanding this, the Applicant/ Licensee shall ensure that there are adequate means of ensuring the quality of the incoming components.